

Dustin Schimek (00:19):

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Brandi Beals (00:40):

Hello everyone and welcome back to the Data Ideas podcast. I am one of your co-hosts, Brandi Beals. Dustin is not able to be with us today, but I'm really excited about the conversation and the guests that we have for you. Today I'm joined by Nicole Mark, an award-winning data visualization engineer and Tableau Ambassador based in Tampa, Florida, as well as Randy Buchman, the founder and CEO of Pleasantries Cannabis Company and a Tableau enthusiast based in metro Detroit. Thank you both so much for being with me today.

Randy Buchman (01:17):

Thank you for having us.

Brandi Beals (01:20):

I always love to start with a bit of background. Obviously I gave you very brief intros, but could you each share what your career journey has been like and how you found a passion for data? Nicole, maybe you can get started.

Nicole Mark (01:34):

Yeah, sure. I have an undergraduate degree in English communications and technology from the University of Maryland [Baltimore County] (UMBC). You can imagine how lucrative that was when I got out of college. So I had been when I was in college working as a pharmacy technician. Then I graduated and I was like, I'm, I told my advisor I was going to go to grad school for English, and she pointed to a pile of folders in the corner and said, those are the applications for one part-time, not tenured English instructor at the college level, and that's what I wanted to teach. So I was like, I don't think I'll do that. So I ended up staying in the world of pharmacy for a bit longer. I worked at the University of Maryland Medical Center for three years where I got certified. I learned how to make IV admixtures. I learned about sterile technique and sterile environments and all this stuff.

(02:43):

I got a job offer from the University of North Carolina in 2011. So I moved down to Chapel Hill in 2011. I was in North Carolina for seven years and I worked at UNC and I worked at Community Care of North Carolina Healthcare, an ACO - a Medicaid administrator - and then I moved to Tampa five years ago to join Aspen RxHealth, a startup here in the digital health space. So my passion for data is lifelong. I just didn't know that initially as a kid. I would keep a little spreadsheet of all my library books and all my Baby-Sitters Club books and all these things. I loved to make maps back then. I like to make maps now, too. So I feel like I always loved data. I liked statistics as a mathematical subject in high school. I did really well at it. I liked anything I was good at, but I was also interested in it. So while I didn't plan a career in data per se, it's not surprising that I ended up in data. The way I ended up in a data, completely data-focused role was after Community Care of North Carolina. I went to work in Asheville at Mission Health and I analyzed the data coming from the pharmacy automation. So we had machines that made sterile products, whatever. So that was my first real data analyst job and the rest is, I discovered Tableau when I was at Aspen, changed my life during the pandemic. All I did was learn Tableau better.

Brandi Beals (04:38):

Yeah,

Nicole Mark ([04:38](#)):

That's

Brandi Beals ([04:39](#)):

Cool. It's, actually, you know, I love your story because it's really not - Sometimes people think that getting a job in data is a very linear path, and I think your story kind of shows it's really, at least for you, it seemed like kind of a lot of little steps, very incremental. You had this job and you stuck with it because this other path just seemed maybe like a constant uphill battle and then

Nicole Mark ([05:08](#)):

I was into probability at the time too. And I was into probability at the time too, and I was like the probability of me getting out of that little stack over there.

Brandi Beals ([05:16](#)):

Yeah, no, but I think that that's really wise, right? But you took that job you had or that closeness you had in pharmaceuticals to get a job in healthcare, and then you got into data and then you kind of just progress and you kind of get deeper into that passion of yours and buy-in. That's really, I love that story. Randy, could you give us a bit of background on your, you know, career progression or just kind of how you found your way to data?

Randy Buchman ([05:48](#)):

I certainly can. I think my path's been more of defined by just a good understanding of math and my career path's kind of been a condition of my times. I graduated from Michigan State in 2008 going into that working economy in '09 and '10, terrible markets across the board. I worked some out of college. I was an auditor at a high-end hotel in Michigan, and then also ran food and beverage at a golf club. Worked my butt off and didn't make very much money, and I was just like, that's not going to cover the bills as far as my student loans, et cetera. Cannabis had passed legislation in 2009 for caregivers, or 2008 in 2009, you could actually get a card as a caregiver or a patient and a caregiver. I started cultivating with four lights in my parents' basement at that time and did not know much about cultivation of cannabis.

([06:53](#)):

I did learn all about the economics of cannabis at Michigan State to a decent level, understood that there was a market there and what the margins looked like and prices, et cetera, and if you could figure how to grow it and you knew how to sell. It went from four lights in a basement to consider currently, I'd say I'd parlayed or kinda redlined maximum growth from that time forward for a very long time, until the current day. Moved into multiple where I was kind of overseeing or consulting on multiple operations at a residential level and then moved into a commercial level in 2013, so it would've been 26, 27 at that time. So cultivating under a caregiver structure in a commercial setting so you could actually go to scale and have indoor control 24, 365 days a year in Michigan of being able to harvest, I think it was 72 plants at that time.

([07:56](#)):

And then as legislation moved forward, we had the medical laws passed in 2016. At that point in my life, I'd done pretty well at iterating a few different commercial facilities and really dialing it in. I was at, I'd say the front edge of understanding the technology and mechanical equipment available that wasn't being used in more of the caregiver setting, but more large ag applications and trying to combine cannabis cultivation to systems that were used in other industries that were commonly used in cannabis at that time

was not being done a ton. So really studying and bringing in different methods to cannabis cultivation. I was able to form the company Pleasantrees, or Emerald Growth Partners, which is my parent company, I think in '17-'18. Then had to learn how to raise capital from the capital markets. Did that raised a substantial amount of cash, I would say over two different rounds, so convertible debt notes, learning just different finance structures.

(09:08):

Built a company, built two massive cultivation centers in Michigan here, and the first one was a little over \$12 million build-out, hired a hundred-some people. Actually, we hired over 200 people during 2020 during COVID, which is from 30 to 230 employees in six months was pretty wild during that time. That's what happens when you build these facilities and they turn on and now you need to have staff to operate 'em. During that entire building and just doing so many different things. It was after say maybe a year and a half, two years of just continually building, building. That was my daily routine and that's what I was chasing. It turned into, okay, well let's run and optimize the things we have, and at that point, I had multiple retail stores. I had a robust wholesale business with multiple different brands and different deal structures that were manufacturing for two different facilities for production and distribution, two different cultivation facilities, and a large amount of just transactions and just any direction you could imagine going on.

(10:17):

It was like, "Whoa, okay, whoa, what do we need to be doing?" Sitting in different meetings, it's like, what should we do with this? What are we doing? Those are all the questions that are repetitively getting asked, and a lot of times, I mean this, you're making me relive a nightmare, but bring it back four years ago where different departments were calling the same thing, different things, and then sitting in a meeting trying to understand what we had of what and saying, okay, well, let's get everything down to, let's call the same things, the same things.

Nicole Mark (10:50):

See, he's a data person at heart, just this thing

Randy Buchman (10:54):

And then getting everybody to call the same thing, the same thing across different platforms. At that time, we had Metrc, which is a compliance software and tracks parent-child life cycles of something through our entire supply chain. On top of that, we had different B2B platforms, retail platforms, having five stores, massive amounts of information we can pull in from different places, and then trying to figure out, once had normalized [the available data], it was like, what is the right information? What's real? People showing up at meetings with three different answers for the same thing now, and it's like, what is the golden answer? I don't want to spend, now we're sitting here talking about this. If we just had the answers, we're talking about execution and well. So people didn't have the answer, so I was required to get the answers or I felt responsible to get the answers.

(11:50):

I have to have the answer, somebody has to have the answer, and I'm the backstop of the company making the final decision. So having as much information that's usable to me and as quickly as possible so decisions can be made. Or at least I can make decisions, and/or the information can be given to decision makers in their different roles became like a quest, I'd say three, four years ago, and it hasn't slowed down. The more I learn, the more there is to do. It's a never ending project, but when you get more and more data, you get better answers as long as you can organize it the right way and have it kinda tell you stories.

Brandi Beals ([12:32](#)):

Yeah, I always refer to, I guess what you're describing sounds a lot like, this endless quest is something that I think a lot about. I'm a very type A person and I like to kind of have a plan, execute, complete the plan and be done and satisfied, but that's not what data is. There's always a new thing, and I always kind of compare it to that children's book *If You Give a Mouse a Cookie* [by Laura Numeroff]. So it's kind of like if you give a mouse a cookie, he's going to ask for a glass of milk. If you give him the milk, he wants a straw. It's just like there's always something, but I love just your career arc. It sounds like you've spent, I mean, first of all, it just sounds like you're very entrepreneurial and you've spent a majority of your time in the cannabis industry and building out this company, it sounds like you've had just absolutely massive growth, which is awesome. Congratulations on that. That's phenomenal. And then now to try to get your arms around optimizing things, dealing with terminology and all of that. It's a massive undertaking for sure.

Randy Buchman ([13:50](#)):

Yeah, it's been a while. It's still, I mean, it's still a fight every day. The market's condensed, just like the economy in general, but cannabis has been taking a beating for the last, call it two years, especially in Michigan, very competitive market. So having your information, being able to make decisions as quickly as possible, like a wrong decision. It cost me six figures easily, quickly. So you're forced to be operate that level.

Nicole Mark ([14:19](#)):

I said data is truly essential in that situation.

Randy Buchman ([14:23](#)):

You can't guess better than math.

Brandi Beals ([14:27](#)):

Right, exactly. So two really interesting stories and backgrounds. Obviously Nicole, you kind of started in the healthcare space. How did you two first connect and start working together?

Nicole Mark ([14:45](#)):

It was kind of random. Well, it wasn't random. I can't remember Rachel's title, someone who works for Randy, Rachel, who worked closely with him. She was frustrated that Randy was so upset he couldn't do what he wanted to do with the data, and Tableau was slow. He didn't realize you can't put all the data in one workbook, all of it from your whole warehouse. No, but anyway, she decided to see which data visualization people Tableau was following, and I guess I think I have a weed viz on my [Instagram] profile. That's probably what it was. But anyway, so she set up a meeting through my Calendly from my website. Like people do sometimes, and I had had such a shitty week, I was like, you know what? I just want to cancel this meeting. I don't don't want to work with any new clients right now. I'm just like, whatever. I'm like, no, that's rude. It's in an hour stopping being an asshole. Take the meeting. I'm like, okay. So I start to look at the invite and I'm like, Pleasantrees, okay, let's find out what that is. I'm like, oh, this is cannabis. I'm like, all right, I'm interested. Because Nicole likes cannabis. Yeah, I live in Florida, but we have a medical marijuana program here and I have a [medical marijuana] card and so does everyone else.

([16:18](#)):

And when do you get to play with data about cannabis? That's, like, legit. You don't, because I've been looking for it for years, three or four years. I started a viz that was supposed to be really complex about

the cannabis laws, but it ended up just being like, here's the states that are legal. Here's the ones that are not. During the course of that meeting, which was kind of long for a client intro meeting, but we hit it off, Rachel's lovely. She was there and I had a very specific idea of "business CEO" in my mind, and I stuck Randy in that box immediately of course. So I was like, okay, CEO dude, I'm listening to what he's saying and I'm thinking he's probably more data savvy than average, but we'll see. Yeah. Oh, that answers the question.

Brandi Beals ([17:40](#)):

And Randy, I guess from your perspective, it sounds like you kind of had Tableau already, started working with it, and needed maybe some data/Tableau help, I guess. How built out was your data infrastructure and your data team, when you brought Nicole on?

Randy Buchman ([18:00](#)):

Me at that point, I had gone through different, had different engineer architects or whatever the different titles are, but different iterations. I think I had what, 1700 different worksheets at that time from probably call it two years of working off with different data sets--a lot of 'em stale--but it was kind of like a junkyard of different information from old data sets and or broken because we switched the API key on this platform or that one, and then you had to unpack all that, clean it up, and I remember vividly still the problem that I was at and I spent two months trying to figure it out. It felt like, and I learned a lot and I self-taught myself the different, the tables and what other information was flowing through the API key [APIs]. I knew what information I needed, I guess what I want to get out of it, so getting it to that level so I can then use it and blend it with different things. I had a decent idea I was trying to combine. At the time that Rachel reached out, she was probably felt terrible because I was sitting up beating my head against the screen trying to figure out

Nicole Mark ([19:22](#)):

How to, she was felt so sorry for you. She's like, "I have to find him someone."

Randy Buchman ([19:27](#)):

Yeah, I had two different platforms that the same products were sold in, but they had different product taxonomies, like naming conventions, but normalizing that or keying that together and then aggregating at the same date so I can tie it to my finances. I couldn't figure out how to get the data aggregation.

Nicole Mark ([19:50](#)):

A [data] problem that everyone has.

Randy Buchman ([19:52](#)):

Yeah, and it's a tough one. Rachel reached out first and now it's been just, I think during my process, I didn't - I never took any formal classes or any computer engineering. I was at Michigan State, I was poli-sci. In the James Madison [College] program for three years and I graduated. I transferred and graduated from hospitality business because I could graduate that semester. Computer science, and just my learning process, has been GPT, YouTube, Google. I have this problem I want to [solve] - how do I do this in Tableau or where should I--, just asking different questions too, looking for different places. I'm somebody that knows a lot about, once they understand, once Nicole was spun up on the structure and what I was looking at and what I was trying to, get out of it, she could- I could tell her the math and we could put together the actual data sets that were at that level and I can use 'em and visualize 'em, Nicole can.

Brandi Beals ([21:02](#)):

So this next question, you've touched on it a bit already, but you've clearly invested in both data tooling and some data folks. Why was it important to you to make that investment?

Randy Buchman ([21:21](#)):

There's again, a lot of platforms in cannabis and the platforms don't necessarily talk to each other, so they're segmented into what they like to, they think their world of, or whatever, their section of my supply chain, or I would just call it the supply chain that they are specialized in. They're looking at, or maybe they're looking at the whole thing, but they're good at only a certain section where we use it, a la carte, for certain sections of what we're looking at. Again, there's no common nomenclature or taxonomy in this industry yet, so everything was in different structure and format and having to put that together. I guess I lost track of the question on that part where I was going with that.

Brandi Beals ([22:12](#)):

You answered it. Yeah. Yeah, no, that's really good. I guess I am kind of curious, and we'll probably get into this a bit more later, but at some point I would love to understand if you had a vision for, was it just a specific question that you knew you wanted to answer using data, or did you have a vision for what you wanted your data infrastructure and ecosystem to look like and you just wanted to move the needle?

Randy Buchman ([22:44](#)):

I think the best, the honest answer and the best answer is, I think three years ago, I kind of drew this out of my whiteboard. I'm a manic maniac and all my partners were looking at me, what is wrong with this guy? I drew out kind of an algorithm and a circular kind of multifaceted calculation. This calculates input output spins back over here, and it's all the answers of what drives every decision at every point in the company. What plant do we put in the ground? What do we turn this into when we harvest it based on information going out the front end of velocities and gross margins and sales through my wholesale retail channels, inventory levels, looking at macro market data. It's always been that. Then how do I do that well, in order for it to be the best answer, it needs to calculate with the most real live data. So I can have as far as the most accurate predictive data telling me what to do going forward. I need to make those decisions. Every Monday we're harvesting three rooms and thousands of clones are getting cut and going in the ground, and I have a five month, four month cycle from when that clone goes in the ground until it's harvested and going into a finished good and finished good channels that going into. So I got to be looking four or five months ahead and be right. And

([24:14](#)):

And to steer from the back. It was like, I guess the one sentence to everybody in cannabis always was trying to sell what they grew, and that was easy when the markets were different, and when cannabis was new and it wasn't saturated like it is, especially in Michigan now. What that what has forced me to do, and I think anybody that's going to be surviving needs to really be focused on flipping that [paradigm] and kind of producing and cultivating what people want, what's selling, with as much information as possible to make the decision of what to do.

Brandi Beals ([24:52](#)):

Yeah, hopefully you took pictures of that whiteboard. I bet it's a moment to remember.

Randy Buchman ([25:00](#)):

It's still so true. It's wild looking at it actually. It's like nostalgia. It's like that's still the fucking answer.

Nicole Mark ([25:05](#)):

One night, we were working late on a Zoom or something and he was like, I've got to find it, and he found it.

Brandi Beals ([25:14](#)):

That's great.

Nicole Mark ([25:15](#)):

No, I think also to watch the evolution of how he tried to apply this, right? So really what, anyway, I'll look at his old Tableau workbooks and I'm like, oh, I see what he's trying to do here, and then what does the next one look like and the next iteration and his naming conventions are pretty consistent, so I can usually tell what's the next thing that he did, or I can find out the date if I want to write [some code].

Brandi Beals ([25:44](#)):

It's like Randy's awesome Tableau workbook. V three version final.

Randy Buchman ([25:50](#)):

Yeah, final Master Gold V two. Yeah, we've all

Nicole Mark ([25:54](#)):

Mag, magnum opus.

Randy Buchman ([25:54](#)):

Audited.

Brandi Beals ([25:55](#)):

I love that. No, and Nicole, I think one thing just to kind of chime in a bit on what you just said, it's great to look back and see one how far we've come see that evolution, but also I think sometimes there's this idea that the thing that we create, whether it's in Tableau or in a database or something, needs to be perfect, and that's never really true because we grow and evolve and we learn more as people, humans and data practitioners alike. And there's a phrase that one of my colleagues used to say, which is that every solution is an interim solution, and it really made me feel good about not necessarily trying to get something to be perfect, but just have it be like progress, right? It's better to have the next good version of something than to try to achieve perfection. Never really true. Also, it gives you kind of freedom to throw something out and to make it again or start again. So yeah, every

Nicole Mark ([27:07](#)):

Solution is, you have no idea how salient your point really is,

Brandi Beals ([27:13](#)):

But we've all been there.

Nicole Mark ([27:15](#)):

I came into this thinking, okay, this is a hard data problem, but I'm going to solve this problem. No problem. I'm going to work hard. I'm smart. I'm going to solve it. It has been so humbling. I'm not even kidding. I'm not even kidding. It's been so humbling for me, and it's not imposter syndrome or something

like that. It's like, no, actually, Nicole, you have this much more growth that you need before you can take this and make magic out of, well, you're never going to do this fast. Many clients I've had in the past, I'll say, do you want it right or do you want it fast? And I know that Randy wants it, right? I mean, he also wants it fast, but right is more important to him than fast.

Brandi Beals ([28:13](#)):

Well, again, I don't mean to make so many references, but I think honestly, you're serving up softballs. So first of all, there's this idea that you can have it, that you can get it fast, accurate, and [cheap] it's like you have to pick two, you have to pick. You can never get all three, you have to pick two. But

Nicole Mark ([28:38](#)):

Also this is, so tooling is getting very expensive. I think you're probably going in a more interesting direction. So I won't say anything else about that other than I haven't had to evaluate so many vendors as I have in this last maybe six months. And I mean, it's crazy,

Brandi Beals ([29:04](#)):

But also I would say data professionals right there, there's a finite number of resources. And depending on, I'm sure where a company is in their data journey, you might be looking for someone who's experienced who can get you up and running, and that I'm sure costs something, right? So there's that element too. But I think the other thing I was just going to talk about in this humbling experience that you mentioned, Nicole, is the Dunning Kruger effect, if you're familiar with that, I love that reference. And honestly, again, you have to be like, I don't know Neil deGrasse Tyson to never at some point find yourself on the left side of the Dunning Kruger effect curve. I

Nicole Mark ([29:58](#)):

Had some unrealistic expectations for myself.

Brandi Beals ([30:02](#)):

We all do.

Nicole Mark ([30:03](#)):

I was like, oh, I can totally be the data engineer, the data analyst, the data visualization designer, and what else? Data ops. I mean, no, I can't do all of those things.

Brandi Beals ([30:16](#)):

So there four distinct jobs,

Nicole Mark ([30:19](#)):

No matter how much I want, I really want to help Randy make his vision real. And I haven't been super passionate about making any CEO visions real before. I've been passionate about making charts, but I'm invested in the business also, not just in the metrics of the business, if that makes sense. A hundred percent. He's such a good business person. He's a decent human, which is hard. I mean, it's just hard these days. It's crazy. And I believe in what he has on the whiteboard. It's just taking some time to get it into programming languages. Tools. Yeah, go ahead.

Randy Buchman ([31:19](#)):

What you just said about the bridges or good answers or the next best answers better than chasing that rabbit. That's been, I guess the last six to eight months of my experience in my data life of I got to a certain spot where I had all the information. Like I said, it's kind of like a multidimensional circular algorithm, and I've been able to chase out all of the different inputs coming from the different data systems to as live as possible, at least what I was able to get via an API into a Snowflake account, and I got the right answers, the right inputs, but then learning that, okay, now I need to go to SQL. Okay, what's that? And then going to, well, okay, some of this stuff is very complex and can't be done in a SQL and needs to be done in code at let's go direct through the APIs, and especially if I want to make it programmatic, learning different levels of what the actual finished product to execute what my vision is, and then making the decision, okay, well, operationally, my team still needs that SQL query output so that they can make these decisions in this department over here.

(32:40):

Or they're getting these reports that they know are true and golden data. Do I do that to or spend time building that visual out here when we know that the finished products here? Well, it's kind of throttling both at the same time. You're building or fixing a plane as you're flying it or changing the tires as you're driving the bus. It's making that decision throttling now. It's tough.

Brandi Beals (33:10):

Oh, I'm sure it is. And it sounds like you have multiple things in the works too, right?

Randy Buchman (33:14):

Yeah.

Brandi Beals (33:15):

Talked a bit about that kind of golden data or that single source of truth that's kind of like a data governance, data quality, data validation type of exercise. But you're also talking about tooling with APIs and Snowflake and SQL, and Nicole, you've mentioned kind of playing multiple roles in terms of data engineer, data visualization, engineer and designer, and all of these different roles. So yeah, you guys have a lot going on all at once. That's a lot to manage. Nicole, one of the things I wanted to ask you was to get your perspective because fairly new to the cannabis industry, but not new to data. And I'm kind of curious to get your perspective on what it's been like to navigate an emerging industry where maybe there's not either a common data framework or maybe even common language around some of these terms. I'm curious to get your take.

Nicole Mark (34:35):

So in some ways, cannabis is like every other industry. They need sales dashboards. They need, how long did it take for a customer to get something to, they need the normal metrics, average order value, gross profit, all these things. In some ways, and also in some ways, as you've already heard, we have the same exact data problems as every other organization to varying degrees. You've heard all the, I think you've got all the greatest hits in there. I mean, I think we covered 'em all, but then, yeah, there's this very cannabis industry-specific data that is challenging, partly because there's the regulatory body, the CRA, and they call one thing, it's the identifier problem, and then we have our wholesale, our wholesale software vendor calls it another thing, and then our POS system calls it something else, and then we want to call it something that makes sense. So then there's another one.

(36:02):

Randy's done a good job of working with our technology partners. He wanted a change in the product hierarchy, and he just kept at it until he got the right person. And CEOs don't do that. They don't have

time for that, but he knows it's important because certain, of course, certain categories are more profitable, less profitable. I want to come back to your question about an emerging part of it is the same for me. I worked at five startups in a row, so it just feels like startup vibes to me, and I still work a lot. I still do a lot of jobs, so I mean, it's a space I'm comfortable in and I learn every day from the data, not just new ways I have to manipulate the 80 columns of data or whatever, but the cannabis data is fascinating, and that's part of our problem is we have not put a roadmap on a Figma board yet, but it's in our minds, I'm giving him [Randy] a hard time, but we want the operational reporting to be perfect so people trust the data, and this business has gotten burned by the data people before, so I don't want to be another one who's not giving them what they need. And then on the other hand, every three days, Randy has some other project that sounds fascinating to me, so then I'm like, I'm don't tell me anymore things that sound fun. Then my ADHD doesn't want to do the day-to-day.

Randy Buchman (38:12):

Yeah. I think,

Brandi Beals (38:13):

Oh, sorry. Go ahead, Randy.

Randy Buchman (38:15):

To piggyback on that, I remember where I left off in the previous response when I trailed off where I was going with it, why I got it, why it became important to my company to have our own internal data team, and why I made the investments with the segmented different platforms. I started to realize probably two and a half years ago, I was able through my technology partners, my different softwares that were using for transactional data and or tracking or even visualization that, or I was able to get meetings with the right people, certainly probably because of my title in my email when 'em requested a meeting or I need this answer, I'm getting through to the top developers widely, and fortunately enough, one of my good friends from college at Michigan State actually was the previously founder and CEO, I believe, of Dutchie, which is my largest or the largest PLS and transaction SaaS inside all of cannabis.

Speaker 5 (39:18):

Yeah.

Randy Buchman (39:19):

So talking to him about some of my ideas and what I wanted to do with the information. Two years ago, I started to gain an idea or a view that, or understanding really at that time that a lot of the people in the technology space weren't looking at or chasing the right answers. And then when I was asking, well, if I need this or how can I do this? I was actually telling them different solutions and answers, and I was like, well, I don't want you to go build it, then you're going to go build it for my competition. That takes away any edge I have on my competition because I'm actually chasing this answer and they're not even asking these questions. That being said, I was able to work with the technology providers to a certain extent where I was able to get the information I needed, give me different columns, or let me change the identifiers so I could macro update things as quickly as possible, or I was always asking for different things that they're just like, "What are you doing with this? Nobody needs this." Asking for new endpoints to be created, et cetera.

(40:25):

Then it was like, well, protecting what I'm doing with this information required me to first of all get my understanding up to a certain level and then have an internal team and make it very important. The best thing that we could do as a company would be to change the base data work and make sure that's being

controlled on the inputs. It's kind of like turn the hose off at the faucet versus mopping up at the end of the hose, controlling that as much as possible, and then setting parameters around it.

Nicole Mark ([40:55](#)):

And right to that point, of course, I've already started working on this. We need an ontology. When I was at Olive, I had the amazing fortune to work with an ontologist. I volunteered to do it. He needed some help or something, and so I got to learn how an ontology is built. So it was very, "See one, do one, teach one." So I've been building an ontology in WebProtege, and Randy still doesn't understand, or maybe he does understand, he still doesn't have time to care about that is what I should, say, right?

Randy Buchman ([41:37](#)):

I've learned a lot about ontology in the last month.

Nicole Mark ([41:40](#)):

I know, but you don't have time to sit with me and let's suss out what this means and what does this mean? And I mean, we would both love to do that, but do you have all your CEO stuff? And it's stupid.

Randy Buchman ([41:57](#)):

I think I do have, I have the ability or the answer to every, most of the decision points in my company of what should be done at this point with this, what should we make? Those are, again, math-driven decisions. So those answers are there, and I know those answers, but how do I put myself in a position where making hundreds of decisions a day at a hundred different decision points throughout the company? I can't, so I can train people. And communications, massively important in companies. However, the more information I can put in front of those people or give 'em the math driven decision, if that's what I'm going to look at, this is how Randy's going to analyze the situation and that solution. Well, they could have that in front of 'em all the time. They can change and go against that, but then I'm going to see that and say, well, why didn't you follow the algorithm or follow the work order that was given to you? And if they don't have a great answer, then it's like, maybe you should follow this. And that's kind of where it goes from there.

Nicole Mark ([43:04](#)):

So even he's wanting to empower everyone with data. CEOs act like that that's what they want to do, but he really does want to do it.

Randy Buchman ([43:19](#)):

So long as I can know that I gave them the right, well, as long as they got the best possible math-driven decision from it. And then you can take an overlay, whatever information is available to you, put parameters on it, but under certain controls, so you can't completely throw the algorithm out of whack or process to do something that's off the parameters. I want people to be able to overlay the information they may have that I can't see, and that can't be written into code, but lemme get you

Nicole Mark ([43:52](#)):

Use their cognitive effort on making meaningful decisions around those edge cases and all of the day-to-day easy decision making is taken care of, agree with the data, whatever that looks like, clicking, submit, whatever.

Brandi Beals ([44:15](#)):

Yeah, I mean, it sounds like a really cool vision. I think it's obviously really valuable and important for a CEO to have such a data-driven mindset and to be putting in place this culture of like, well, we want to have data drive some of these decisions, or we want to have that information and being able to collect it and use it effectively. Obviously, it's the dream I think, of every data professional to work with somebody like that. So green

Randy Buchman ([44:49](#)):

Looking to hire geniuses.

Brandi Beals ([44:51](#)):

Yeah, I love that. That's great.

Nicole Mark ([44:57](#)):

No, I mean that's why by the end of one meeting with him, I was like, oh, I want to work with him. He actually does want to do some stuff with data,

Brandi Beals ([45:09](#)):

And that's really cool. And I think the other thing I was just going to say is I think being in a newer industry, I don't know whether that is a accurate statement or assessment, but I guess I think for a lot of people, cannabis is new. I mean, I don't think...

Randy Buchman ([45:28](#)):

...is nascent,

Brandi Beals ([45:30](#)):

It's legal in Wisconsin in any form, not, I guess I, I don't know whether Randy, you would say that it's kind of emerging. Obviously it sounds like it's been established in Michigan for quite a long time, but I think if you're thinking about expanding to other states anyway, I guess I just think maybe at a national level, it's an emerging industry, and I can imagine that there's not as many data providers as there are in other industries that have been established for hundreds of years. And so while it seems like you're dealing with questions of an ontology and some of these things that you kind of are having to figure out for the first time, on the other hand, Randy, you can go to the folks where you're using their API and saying, I want these additional fields, you're also able to influence things I think a lot more too. That's what it sounds like to me. I'm curious whether that's accurate.

Randy Buchman ([46:41](#)):

Yeah, absolutely. I definitely demand certain things. We're a substantial or a decent sized business where I could, granted, I try not to get locked into softwares, not contractually, but just where the lift of leaving or transitioning out, it's so painful that it's less painful to not have that caliber quality of service and or prices, et cetera. But I'm not scared to drop any platform I don't think and move, since we do have pretty clean data, it's very easy for me to transition. So when I say, listen, I need this and this, or I'm going to go to another platform that will give me that and that because what I need to operate my business, I'm pretty successful in getting results. And my asks are usually, I'm asking, I already tried to get it six other ways without having to ask you. So I understand exactly what I'm asking for, and there's usually no good reason why they shouldn't provide me with that information.

Brandi Beals ([47:45](#)):

I just think it's an incredibly powerful position to be in, to be able to lay a lot of that groundwork for future, whether it's, I mean, of course they'd be your competitors, but for other people who would also be essentially building businesses based on a lot of the work that you've done, as I'm sure that it doesn't maybe sound exciting from a CEO's position when the market, obviously again, they'd be your competitors, but I think from just making a name for yourself and for your company and to be the leaders in that space, I think, I mean, it's incredibly exciting.

Nicole Mark ([48:29](#)):

I would've been dumb. No, Brandi, you can see all of why I am doing this. Yeah, go ahead. Randy, you were going to say something? I'm sorry.

Randy Buchman ([48:44](#)):

Yeah, I think it's a noble cause. I think there's some beauty to the data as well. Unlike what's unique to a lot of industry is I discovered early in my trying to track and identify everything through my supply chain and talking to some friends that had logistics software my buddies had, his family had owned a massive logistics company and had a pretty serious piece of software from the nineties or whatever, that was still extraordinarily valuable, and it tracked inventory down to, it was able to go down to a pallet in the back of a warehouse anywhere in the country level and in cannabis, we're tracking to an ID per unit, like a unique identifier. The beauty of the data is that there's an answer in every single thing that's out, and if it doesn't, that's an outlier. It has to,

Nicole Mark ([49:38](#)):

Yeah, it's most likely a problem. If you cannot find that

Randy Buchman ([49:46](#)):

The money always has to net out with the product. So if the product can always zeroes out, I can track every dollar, every penny. So there's a real beauty to that and knowing that I know what I'm chasing, so if I know my costs and everything in real time, I can know where the profit is in as real time as possible, and then I can forecast and calculate and drive the business that way.

Nicole Mark ([50:12](#)):

Of course, having worked at UNC and with researchers in my consulting, I also think about the science. He has very sensitive equipment that gathers these environmental pieces of equipment, gather data 24 7. Yeah,

Randy Buchman ([50:32](#)):

I guess I didn't really discuss that that much is the one thing I think

Nicole Mark ([50:35](#)):

That's the most fascinating. But

Randy Buchman ([50:38](#)):

Yeah, one thing, even before, I guess my data journey started before I did because I did it early in the business even when designing my facilities. So pre going through that explosive growth period and running around developing and building and chasing real estate and licenses before I settled down into optimization, I still had the foresight to make sure I was collecting as much data everywhere from having the cultivation background and growing for a decade of my life. I knew what to track and what strains,

yields, environmental data that were driving it. Automations everywhere is possible. When I built my grows, I built spaceships. I have some of the most advanced mechanical equipment, fully automated custom irrigation systems that track what's going into every plant, what's coming out, all my dry backs, environmental snapshots from tons of different places, and I've been grabbing that for five years now, and it's clean, it's organized, and just knowing that I'm going to come back and this is what's going to ultimately help me crop steer. We've been terpene testing on every single thing since day one, so collecting terpene profiles from probably, I don't even know how many phenos we've grown, hundreds or thousands of different terpenes. Is it hundreds? Did you combine both sites?

Nicole Mark ([52:03](#)):

Huh?

Randy Buchman ([52:04](#)):

Combine both sites. I bet you it's probably closer.

Nicole Mark ([52:07](#)):

It's almost a thousand. So I think that's the right number.

Randy Buchman ([52:11](#)):

So having that, I'll put genetic data and having, I have a substantial seed collection, and I spent tremendous amounts of money just compiling over years. We can get the stuff you can do with, it's really fun, but I knew that that would ultimately be driving different parts of decision of what to grow, et cetera, et cetera in the future, and bringing that in and combining that. It's been fun.

Nicole Mark ([52:41](#)):

Yeah, my geeky self gets, I am the most fascinated by that kind of data.

Randy Buchman ([52:46](#)):

I still have piles of data that I know.

Nicole Mark ([52:49](#)):

I know at one point I was just like, don't show me any more interesting things. It's just not, I have to focus.

Randy Buchman ([52:58](#)):

Well, I get to a granular level enough to operate, and then I still know here's the rest of the data to go four levels deeper, but I don't need that right now. I'll come back to it.

Brandi Beals ([53:08](#)):

I mean, that's amazing. And Nicole, I know you kind of joke about like, oh, there's so many awesome, exciting projects to work on, but there are no, but I guess I'm with you. It's like there's no end to the things that I want to do,

([53:30](#)):

But obviously for my own mental sanity, I need to keep that number realistic and finite. But it's also got to be just really exciting to work in a space where you're excited about the company. You obviously get along really well with the CEO who champions the work of data professionals and being able to glean

insights from data and just getting excited about, I mean, Randy's sitting there collecting his own data from the very beginning. I mean, that's brilliant, and it's just got to be really exciting to work with somebody like that, but also in a space where there are just clear skies and just open waters of amazing data projects to pursue. I think that's phenomenal.

Nicole Mark ([54:24](#)):

I mean, all I can think of is the papers we're going to publish one day, that's in my mind,

Brandi Beals ([54:31](#)):

You're a nerd.

Nicole Mark ([54:32](#)):

He's thinking about the money and I'm thinking about the academic notoriety.

Brandi Beals ([54:36](#)):

Yeah, I love that. That's great though. I love that. Cool. Well, we've been talking for almost an hour now, so keeping us all on track with our dates. I'm going to just kind of maybe ask you the question that I always love to ask at the end, which is, how can people find out more about you and the work that you do?

Randy Buchman ([55:00](#)):

Um,

Brandi Beals ([55:00](#)):

Randy, do you want to go first?

Randy Buchman ([55:01](#)):

Our website is www.enjoypleasantrees.com. Instagram is [@wearepleasantrees](#). I think we got some really cool things going on, especially in the data world and using our data to do some really cool things that I think are disruptive and unique to any industry, certainly to cannabis.

Brandi Beals ([55:25](#)):

That's awesome. Thanks. Nicole, what about you?

Nicole Mark ([55:29](#)):

Oh yeah, Women in Dataviz [Community] Slack. I run that thing. Let's see. Let's see.

Brandi Beals ([55:42](#)):

And you do virtual coffees right? Through the Women in Dataviz?

Nicole Mark ([55:45](#)):

Yeah, next Thursday. Next Thursday morning at eight and the following Wednesday at noon. Women in Dataviz's URL is campsite.bio/womenindataviz, and that has all the links to the things that we do. And there's a link to my website there too. Nice. But I'm easy to find.

Brandi Beals ([56:11](#)):

Yeah, you are. Just Nicole Mark on LinkedIn, she'll show up. Yeah, it's great. Cool. Well, thank you both so much for joining me today. It's been really exciting to talk with you about this new industry and your data journey. I've definitely enjoyed myself, so thank you both so much for your time today.

Randy Buchman ([56:32](#)):

Thank you for having us.

Brandi Beals ([56:33](#)):

Yeah,

Nicole Mark ([56:34](#)):

It was fun.

Brandi Beals ([56:35](#)):

Absolutely. Yeah. And for all of our listeners, don't forget to like this episode. Rate the podcast and share this episode with friends. It really helps us out a lot. Thank you all, and have a great day.